

KOFvision

KOFvision is a standalone Python viewer for KOF survey files with:

- 2D and 3D visualization
- KOF polyline parsing (09 91 ... 09 99 / 09 96)
- surface triangulation (Delaunay)
- cut/fill volume calculation between two surfaces
- interactive section drawing and profile charts

The main app is implemented in `kof_viewer.py`.

Features

- Open one or more `.kof` files
- Plot points/polylines in 2D plan view
- Plot points/polylines in 3D with optional triangulated surface
- Toggle labels, markers, lines, triangulation, and equal aspect mode
- Select top and bottom surfaces and compute:
 - net volume
 - cut volume
 - fill volume
 - sampled overlap area
- Draw one or more section lines in 2D and open section windows with:
 - top and bottom profiles
 - shaded cut/fill regions
 - net/cut/fill cross-section area

Requirements

- Python 3.10+
- `numpy`
- `matplotlib`
- `scipy`
- `tkinter` (usually included with standard Python installers)

Install Python dependencies:

```
pip install numpy matplotlib scipy
```

Run

Start empty and load files from the UI:

```
python kof_viewer.py
```

Or preload one or more KOF files:

```
python kof_viewer.py "12_A (1).kof" "12_A(1) (1).kof"
```

Workflow

1. Open two surface files (for example existing top and design/bottom).
2. Select Top and Bottom in the second toolbar row.
3. Optionally enable Triangulation.
4. Click Compute volume.
5. Read net/cut/fill in the toolbar summary.
6. Click Draw section and pick two points in 2D to open a section profile window.

Boundary Tightness

The Boundary tightness control filters long triangulation edges (relative to the median edge length).

- Lower values: tighter concave boundary (drops more outer triangles)
- Higher values: looser boundary (keeps more triangles)

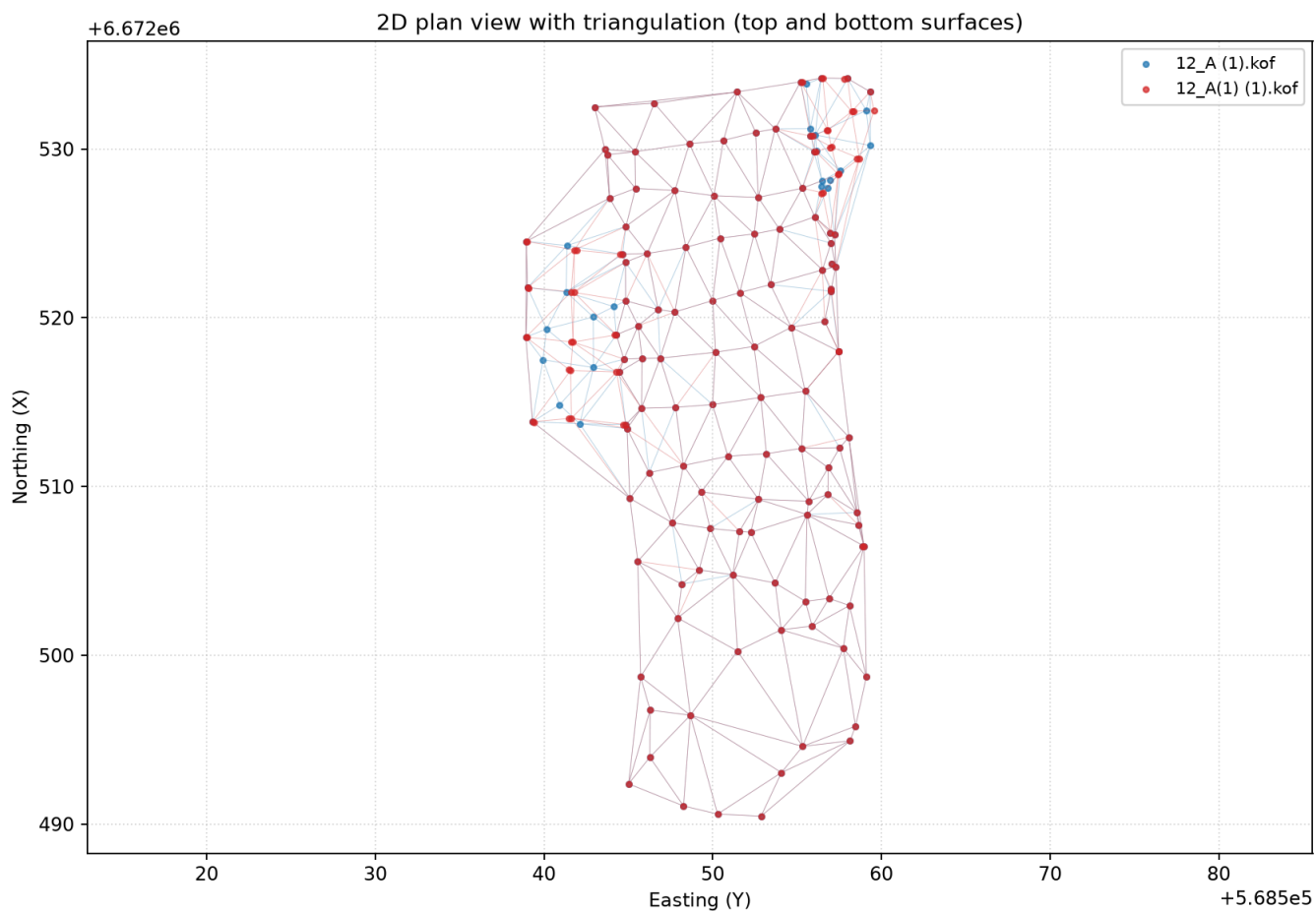
If you change this value after computing volume, the app marks the result as stale and you should recompute.

Example Images

These images are generated from the included sample files:

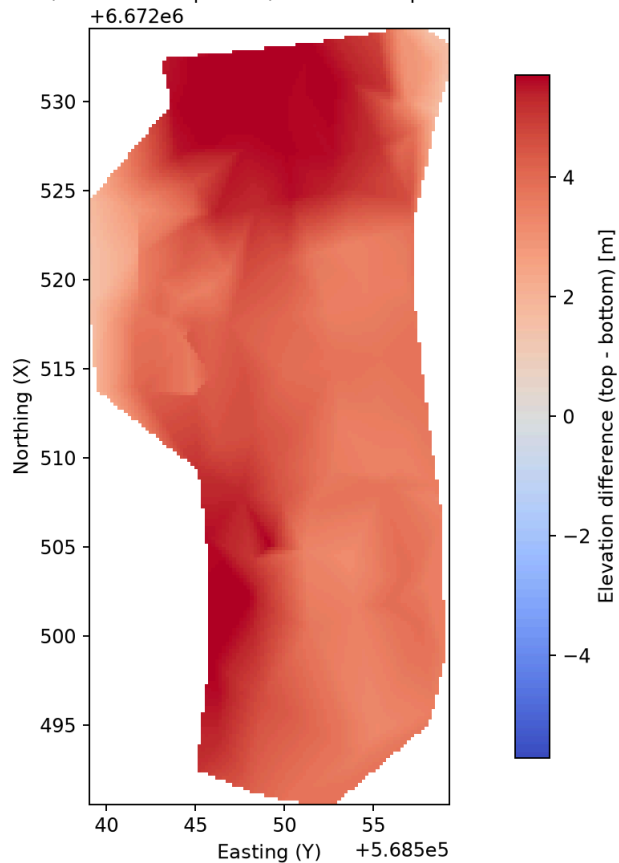
- 12_A (1).kof
- 12_A(1) (1).kof

2D Plan + Triangulation

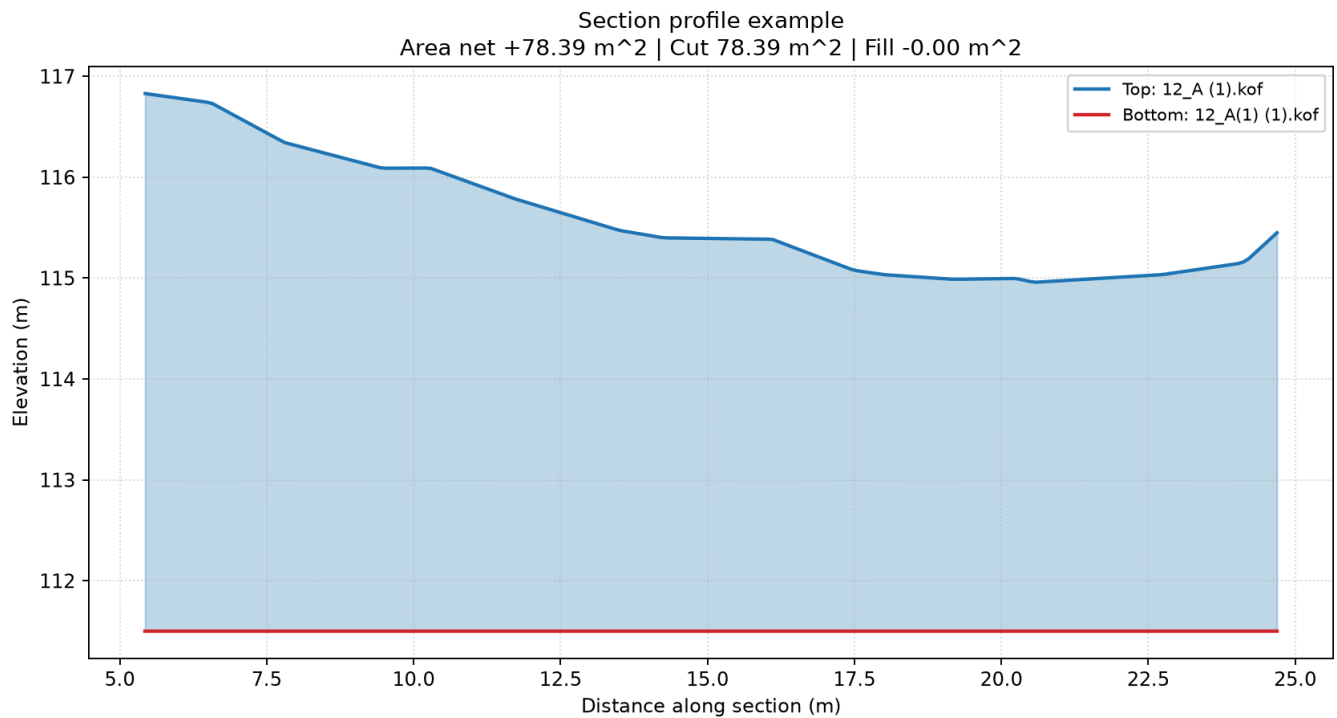


Volume Sampling (Cut/Fill Map)

Volume sampling grid (cut/fill map)
Net +2,742.9 m³ | Cut 2,742.9 m³ | Fill -0.0 m³



Section Profile Example



Regenerate README Images

A helper script is included to regenerate the images:

```
python scripts/generate_readme_images.py
```

It writes PNG files to `docs/images/`.

Notes

- Coordinates are interpreted as:
 - `X` = Northing
 - `Y` = Easting
 - `H` = Elevation
- Volume is estimated by grid sampling over overlapping extents of the two triangulated surfaces.
- Interpolation outside the filtered triangulation boundary is ignored.